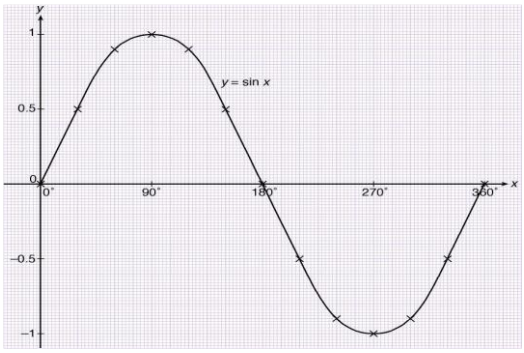
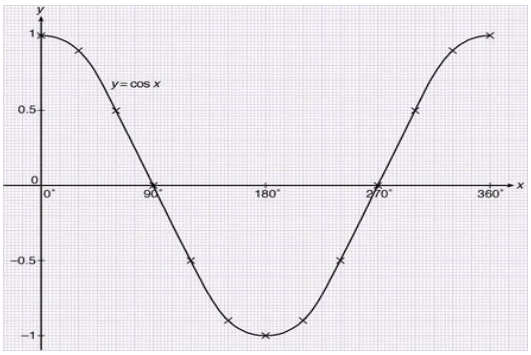
Subsidiary Angle (輔助角)

Graph of $y = \sin x$

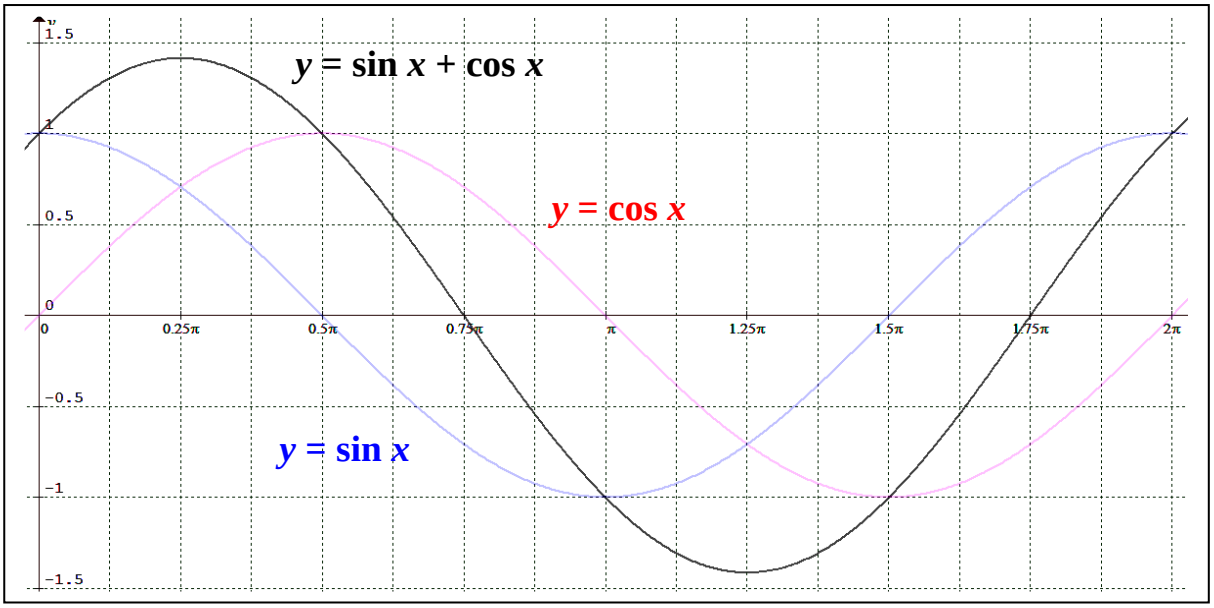


Graph of $y = \cos x$



If it is possible to add the two functions together…

Graph of $y = \sin x + \cos x$



Actually, we can rewrite $y = \sin x + \cos x$ into $y = \sqrt{2}\sin(x + 45°)$!

How to do so? Let's go through it on next page.

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Subsidiary Angle (輔助角)

e.g. By expressing $\sin\theta + \cos\theta$ in the form $R\sin(\theta + \alpha)$, find the solution of $\sin\theta + \cos\theta = \sqrt{2}$.

Solution

Consider

$$\sin\theta + \cos\theta = R\sin(\theta + \alpha) \quad \ldots\ldots (*)$$
$$= R\sin\theta\cos\alpha + R\cos\theta\sin\alpha$$
$$= R\cos\alpha\sin\theta + R\sin\alpha\cos\theta$$

Compounded angle formula:
$\sin(\alpha + \beta)$
$= \sin\alpha\cos\beta + \cos\alpha\sin\beta$

By comparing coefficient,

$$R\cos\alpha = 1 \quad \ldots\ldots (1)$$
$$R\sin\alpha = 1 \quad \ldots\ldots (2)$$

$(1)^2 + (2)^2$: $R^2\cos^2\alpha + R^2\sin^2\alpha = 1^2 + 1^2$

$$R^2(\sin^2\alpha + \cos^2\alpha) = 2$$
$$R^2(1) = 2$$
$$R = \sqrt{2}$$

$\sin^2\theta + \cos^2\theta = 1$

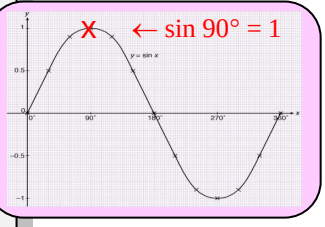
$(2) \div (1)$: $\dfrac{R\sin\alpha}{R\cos\alpha} = \dfrac{1}{1}$

$$\tan\alpha = 1$$
$$\alpha = 45°$$

$\tan\theta = \dfrac{\sin\theta}{\cos\theta}$

From (*),

$$\sqrt{2}\sin(\theta + 45°) = \sqrt{2}$$
$$\sin(\theta + 45°) = 1$$
$$\theta + 45° = 90°$$
$$\theta = 45°$$



The above solution is well known as subsidiary angle method.

Here are general formulas for subsidiary angle:

If

- $A\sin\theta + B\cos\theta = R\sin(\theta + \alpha)$
- $A\sin\theta - B\cos\theta = R\sin(\theta - \alpha)$
- $A\cos\theta + B\sin\theta = R\cos(\theta - \alpha)$
- $A\cos\theta - B\sin\theta = R\cos(\theta + \alpha)$

then $R = \sqrt{A^2 + B^2}$ and $\alpha = \tan^{-1}\dfrac{B}{A}$, where $R > 0$ and $0° \le \alpha \le 90°$

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HKCEE Questions (level 2)

1. ***[A.Math 1993 Paper II Q.2 Modified]***

By expressing $\sqrt{3}\cos x - \sin x$ in the form $r\cos(x + \alpha)$, find the solution of the equation $\sqrt{3}\cos x - \sin x = 1$ for $0 \le x < 2\pi$.

(5 marks)

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HKCEE Questions (level 2)

2. ***[A.Math 1997 Paper II Q.4 Modified]***

By expressing $6\sin x + 8\cos x$ in the form $r\sin(x + \alpha)$, find the solution of the equation $6\sin x + 8\cos x = 5$ for $0° \le x \le 360°$. (Give your answer correct to the nearest degree.)

(5 marks)

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HKCEE Questions (level 2)

3. ***[A.Math 1981 Paper II Q.4 Modified]***

If $12\cos 3x - 5\sin 3x = r\cos(3x + \theta)$, where $r > 0$ and $0° < \theta < 90°$, find r and θ.

Hence find the solution of the equation $12\cos 3x - 5\sin 3x = 13$ for $0° \le x \le 360°$.

(Give your answer correct to the nearest degree.)

(6 marks)

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HKCEE Questions (level 2)

4. ***[A.Math 2001 Q.6 Modified]***

(a) If $\sin x + \cos x = r\cos(x + \theta)$ for all values of x, where $r > 0$ and $-\pi < \theta \leq \pi$. find the values of r and θ.

(b) Find the solution of the equation $\sin x + \cos x = \sqrt{2}$ for $0 \leq x < 2\pi$.

(5 marks)

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HKCEE Questions (level 2)

5. ***[A.Math 2006 Q.9 Modified]***

(a) Express $\cos\theta - \sqrt{3}\sin\theta$ in the form $r\cos(\theta + \alpha)$, where $r > 0$ and $0° < \alpha < 90°$.

(b) Find the solution of the equation $\cos 2x - \sqrt{3}\sin 2x = 1$ for $0° \le x \le 360°$.

(6 marks)

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HKCEE Questions (level 2)

6. ***[A.Math 1989 Paper II Q.5]***

Let $y = 5\sin\theta - 12\cos\theta + 7$

(a) Express y in the form $r\sin(\theta - \alpha) + p$, where r, α and p are constant and $0° \leq \alpha \leq 90°$.

(b) Using the result of (a), find the least values of y.

(5 marks)

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HKCEE Questions (level 2)

7. ***[A.Math 1990 Paper II Q.6]***

(a) If $\cos\theta + \sqrt{3}\sin\theta = r\cos(\theta - \alpha)$, where $r > 0$ and $0° \le \alpha \le 90°$, find r and α.

(b) Let $x = \dfrac{1}{\cos\theta + \sqrt{3}\sin\theta + 5}$, find the range of values of x.

(5 marks)

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HKCEE Questions (level 3)

8. ***[A.Math 1987 Paper II Q.12]***

(a) (i) If $7\sin\theta - 24\cos\theta$ is expressed in the form $r\sin(\theta - A)$, where $r > 0$ and $0° \le A \le 90°$, find r and A.

(ii) Let $y = 14\sin\theta - 48\cos\theta + 14$.
Using the result in (i), find the maximum and minimum values of y.
Find also the value of θ at which y attains its maximum, where $0° \le \theta \le 90°$.

(12 marks)

(b) α and β are two acute angles satisfying the equation $6\cos^2\theta - 5\cos\theta + 1 = 0$
Without solving the equation, show that

$$\cos\frac{\alpha+\beta}{2} + \cos\frac{\alpha-\beta}{2} = \sqrt{2}\,.$$

[Hint : $\cos^2\frac{x}{2} = \frac{1}{2}(1 + \cos x)$.] ***(8 marks)***

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HKCEE Questions (level 3)

8. *[A.Math 1987 Paper II Q.12]* *(Cont'd)*

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Answers

1. $\frac{\pi}{6}$ or $\frac{3\pi}{2}$

2. **97° , 337°**

3. **112° , 232°**

4.(a) $\sqrt{2}$, $-\frac{\pi}{4}$

(b) $\frac{\pi}{4}$

5.(a) **$2\cos(\theta + 60°)$**

(b) **0°, 120° , 180° , 300° , 360°**

6.(a) **$13\sin(\theta - 67.4°) + 7$**

(b) **–6**

7.(a) **2, 60°**

(b) $\frac{1}{7} \le x \le \frac{1}{3}$

8.(a) **(i)** **25, 73.7°**

(ii) **163.7°**

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